26:1 Micro Harmonic Drive - Overview & Instructions

Product Description

The 26:1 Micro Harmonic Drive is a compact, high-precision gear reduction solution designed for desktop robotics, hobbyist mechanical builds, and precision motion control applications. This strain wave gearing system delivers exceptional positioning accuracy with near-zero backlash, outstanding torque capacity, and virtually vibration-free operation. Perfect for applications where ultra-smooth movement and precise control are essential in a compact form factor.

What's Included

Standard Package (STL Files)

- Flex Splines:
- 6 size variations:
- 1 standard
- 5 larger splines (+0.05mm to +0.25mm in 0.05mm steps)
- Housing Components: All structural elements
- Output Components: Output and related hardware
- Pre-Oriented STL Files: For optimal 3D print performance
- Bill of Materials (BOM): Complete hardware list

Premium Package (STEP Files) Includes everything in the Standard Package, plus:

- STEP Files: For CAD modification
- Complete Assembly Model: For reference (hardware not included)

Key Features

- Reduction Ratio: 26:1
- Form Factor: Compact and desktop-ready
- Precision Fit: 6 flex spline variations for ideal tolerance
- Smooth Motion: Reduced vibration via wave motion
- Efficient Design: High torque in a small package

Warranty & Disclaimer

This is a DIY assembly product. Sweep Dynamics provides files as-is. Final performance depends on print quality, fit, lubrication, and assembly precision. Please test thoroughly before using in critical applications.

3D Printing Settings Table

Parameter	Value	Notes
Layer Height	0.20 mm	Balance of detail and strength
Initial Layer Height	0.25 mm	Improves bed adhesion
Wall Count	3-4	Increases structural integrity
Top/Bottom Layers	7 / 7	Ensures solid surfaces
Infill Density	25-30%	Sufficient for mechanical loads
Infill Pattern	Gyroid or Grid	Strength-to-weight ratio
Infill/Wall Overlap	15%	Enhances bonding
Sparse Infill Area	Min. 15 mm^2	Prevents weak spots
Support Structure	Yes	Required for overhangs
Support Density	15%	Material-efficient
Interface Layers	2	Smooth contact surface
Z Distance	0.2 mm	Clean support separation

TPU Filament Printing Guide

Parameter	Value	Notes
Print Speed	0.20 mm	Balance of detail and strength
Infill Density	100%	Solid part
Support Structure	No	No overhangs

Size Selection Guide

Type	Tolerance	Use When...
Standard	0.00 mm	Start here for most prints
Oversized	+0.05 to +0.25 mm	If fit is too loose